

CURRICULUM MODULE | 2026

GRADES 5 AND 6

Guided experiments, fully in class.

First semester | OAS State Science Day pathway



GRADE BAND PHILOSOPHY

Students in grades 5 and 6 are introduced to the scientific process through a curated set of pre-approved experiments. They experience genuine inquiry in the early stages, then choose from a menu of projects that can be completed entirely at school with materials the teacher provides. This removes the barrier of parental involvement and ensures equity, so every student can participate regardless of what is available at home.

The first nine weeks walk all students through the process together. In the second nine weeks, teachers identify students ready for independent competition entry at OAS and provide additional support to that group. All students complete a simplified display board as a class project. Selected students complete a full OAS-eligible project.

EXPERIMENT MENU

EXPERIMENT	SUBJECT AREA	WHAT STUDENTS MEASURE	CROSS-CURRICULAR CONNECTION
Does water temperature affect how fast sugar dissolves?	Physical Science / Chemistry	Time to dissolve in seconds; grams of sugar	Math: graphing time against temperature. Art: diagram of molecular motion
Which paper towel brand absorbs the most water?	Physical Science / Engineering	Volume absorbed in mL; cost per unit	Math: unit rate and cost comparison. ELA: consumer report writing
Does the color of light affect plant growth?	Life Science / Biology	Plant height in cm over two weeks; leaf count	Art: botanical sketching of weekly growth. History: photosynthesis discovery timeline
Does music tempo affect resting heart rate?	Health / Life Science / PE	Heart rate in BPM before and after listening	PE: heart rate measurement. ELA: connecting to wellness research
Which surface produces the most friction for a sliding object?	Physical Science	Distance traveled in cm; number of trials	Math: averaging multiple trials. Art: diagram of surface textures
Does soil type affect how fast water drains?	Earth Science / Environmental	Drainage time in seconds per soil type	History: soil and agriculture in your region. Art: soil sample display
How does the amount of air in a ball affect bounce height?	Physics / PE	Bounce height in cm at different PSI values	PE: equipment science. Math: graphing PSI against height

SEMESTER 1 TIMELINE

Adjust the start date to your school calendar. Build backward from your fair registration deadline, not forward from the first day of school.

WEEKS	PHASE	PRIMARY CLASSROOM	WHAT STUDENTS DO
1 to 2	Curiosity and Question	Science	Brainstorm interests, explore the experiment menu, select a project, begin the journal
3	Background Research	ELA / History	Read one article on the topic, write a summary, note one real-world connection
4	Hypothesis and Design	Science	Write a hypothesis, identify variables, complete a simple experimental design form
5	Materials and Safety	Science	List materials, review safety rules, set up the data table in the journal
6 to 8	Data Collection	Science, in class	Run the experiment with teacher guidance; minimum three trials; record in the journal
9	Data Analysis	Math	Calculate averages, construct a bar or line graph, describe the pattern in writing
10	Conclusion and Report	ELA	Write a conclusion paragraph using CER; begin a short written report
11	Display Board	Art / Science	Design and assemble the board using the FairGame Display Board Template
12	Presentation Practice	ELA / Science	Practice a 2 to 3 minute oral explanation; receive peer and teacher feedback
13 and beyond	OAS Prep, selected students	Science / ELA	Complete full OAS registration paperwork with the teacher; prepare for the regional fair

JOURNAL CHECKS

Collect journals at the end of weeks 3, 6, 9, and 12. Announce all four dates on day one. The journal stays at school between check-in dates.

TEACHER NOTES

Parent involvement is intentionally minimal

All materials are provided by the school. Students do not need to bring anything from home. The journal stays at school between check-in dates.

The menu is not a limitation

Students genuinely choose within the menu. Their choice of surface type, plant species, or music genre within a category gives real ownership without creating equity gaps.

Physical education connection

The ball bounce and heart rate experiments connect directly to PE class. Coordinate with the PE teacher for heart rate measurement and equipment access.

Art connection

Display board design is a graded art assignment in schools that have adopted this curriculum. Students apply layout, color, and typography from art class. The template provides structure; students make the aesthetic decisions.

Wiggle room

Plan for one experiment session to be lost to scheduling conflicts. Three data collection days instead of four still produces enough trials for a valid result on any menu item.

STANDARDS ADDRESSED

OHIO LEARNING STANDARDS FOR SCIENCE, OLS 2018, GRADES 5 TO 8 INQUIRY BAND

- Identify questions that can be answered through scientific investigations
- Design and conduct a scientific investigation with appropriate controls
- Use mathematics and measurement tools to gather and record data
- Analyze and interpret data to develop explanations
- Communicate scientific procedures and explanations

CCSS ELA AND LITERACY IN SCIENCE, GRADES 6 TO 8

- RST.6–8.1: cite textual evidence to support analysis of science and technical texts
- RST.6–8.3: follow precisely a multistep procedure when carrying out experiments
- WHST.6–8.2: write informative and explanatory texts including narration of scientific procedures
- WHST.6–8.4: produce clear writing appropriate to task, purpose, and audience
- SL.6–8.4: present claims and findings in a focused, coherent manner

CCSS MATHEMATICS, GRADES 6 TO 7

- 6.SP.A.1: recognize a statistical question as one anticipating variability in data
- 6.SP.B.4: display numerical data in plots including dot plots, bar graphs, line graphs
- 6.SP.B.5: summarize numerical data sets in context using mean and range
- 7.SP.A.1 to 2: use random sampling and draw inferences from data